

Classical Probability

- Random Experiment: Outcome can't be predicted with certainty but a collection of every possible outcome can be described before its performance.
 - Ex.1: Toss a coin
 - Ex.2: Cast one red die and one white die
- Sample space: Collection of all possible outcomes.
- Notation: $\mathcal{C}$ —sample space; c —one element; C —collection of the elements (Event).
- Ex.3: In Ex.2, let C be the collection of the pair for which the sum is 7. What is C and what is the probability of event C .

- Set: a collection of the objects. $C_1 = \{x : 0 \leq x \leq 1\}$;
 $C_2 = \{(x, y) : 0 \leq x + y < 3\}$.
- Element of the set or Point in the set: $\frac{1}{4} \in C_1$; $(1, 1) \in C_2$.
- Null Set: $\emptyset$.
- Subset: $C_1 \subset C_2$ if for any $c \in C_1$, $c \in C_2$.
- $C_1 = \{x : 0 \leq x \leq 1\}$, $C_2 = \{x : -1 \leq x < 4\}$, $C_1 \subset C_2$.
- Set operations: Union, intersection, complement

- Point Functions: $f(x) = 3x$; $g(x, y) = e^{-x-y}$.
- Set Functions: function that is evaluated, not at a point, but for an entire set of points.
- Ex. Let C be a set in 1-d space and let $Q(C)$ be equal to the number of positive integers in C . Then $Q(C)$ is a set function. If $C = \{x : 0 < x < 5\}$, $Q(C) = 4$; If $C = \{-2, -1\}$, $Q(C) = 0$.
- Some Notations:
 - $\int_C f(x)dx$; $\int \int_C g(x, y)dx dy$
 - $\sum_C f(x)$: sum extended over all $x \in C$; $\sum \sum_C g(x, y)$
- Ex. Let C be a 1-d set and let $Q(C) = \int_C e^{-x} dx$. Thus, if $C = \{x : 0 \leq x < \infty\}$, $Q(C) = \int_0^\infty e^{-x} dx = 1$.

- Probability Set Function: Let $\mathcal{C}$ be a sample space and let $\mathcal{B}$ be a collection of the subsets of $\mathcal{C}$. $\mathbb{P}$ is a probability set function if
 1. $\mathbb{P}(C) \geq 0$ for all $C \in \mathcal{B}$.
 2. $\mathbb{P}(\mathcal{C}) = 1$.
 3. If $\{C_n\}$ is a sequence of sets in $\mathcal{B}$ and $C_m \cap C_n = \emptyset$ for all $m \neq n$, then

$$\mathbb{P}(\cup_{i=1}^{\infty} C_n) = \sum_{i=1}^{\infty} \mathbb{P}(C_n).$$

- A probability set function tells us how the probability is distributed over the set of events. We will often drop the word set and refer $\mathbb{P}$ as a probability function.

- Theorem 1.3.1. For each event $C \in \mathcal{B}$, $\mathbb{P}(C) = 1 - \mathbb{P}(C^c)$.
- Theorem 1.3.2. $\mathbb{P}(\emptyset) = 0$.
- Theorem 1.3.3. If $C_1 \subset C_2$, $\mathbb{P}(C_1) \leq \mathbb{P}(C_2)$.
- Theorem 1.3.4. For each $C \in \mathcal{B}$, $0 \leq \mathbb{P}(C) \leq 1$.
- Theorem 1.3.5. $\mathbb{P}(C_1 \cup C_2) = \mathbb{P}(C_1) + \mathbb{P}(C_2) - \mathbb{P}(C_1 \cap C_2)$

- Conditional Probability: of event C_2 given the event C_1 , provided that $\mathbb{P}(C_1) > 0$,

$$\mathbb{P}(C_2|C_1) = \frac{\mathbb{P}(C_1 \cap C_2)}{\mathbb{P}(C_1)}$$

- Properties:
 - $\mathbb{P}(C_2|C_1) \geq 0$
 - $\mathbb{P}(C_2 \cup C_3 \cup \dots | C_1) = \mathbb{P}(C_2|C_1) + \mathbb{P}(C_3|C_1) + \dots$, provided that $C_2, C_3, \dots$ are mutually exclusive.
 - $\mathbb{P}(C_1|C_1) = 1$
- 5 cards from 52 cards: conditional probability of an all spade hand (C_2) given that there are at least 4 spades in hand (C_1),

$$\begin{aligned}\mathbb{P}(C_2|C_1) &= \frac{\mathbb{P}(C_1 \cap C_2)}{\mathbb{P}(C_1)} = \frac{\mathbb{P}(C_2)}{\mathbb{P}(C_1)} = \frac{\binom{13}{5} / \binom{52}{5}}{[(\binom{13}{4} \binom{39}{1}) + \binom{13}{5}] / \binom{52}{5}} \\ &= 0.0441.\end{aligned}$$

- Multiplication Rule: $\mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2|C_1)$
- 52 cards: Compute the probability that the third spade appears on the sixth draw. C_1 —the event of two spades in the first five draws; C_2 —the event of a spade on the sixth draw; we want $\mathbb{P}(C_1 \cap C_2)$:

$$\mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2|C_1) = \frac{\binom{13}{2}\binom{39}{3}}{\binom{52}{5}} \times \frac{11}{47} = 0.0642$$

- $\mathbb{P}(C_1 \cap C_2 \cap C_3) = \mathbb{P}((C_1 \cap C_2) \cap C_3) = \mathbb{P}(C_1 \cap C_2)\mathbb{P}(C_3|C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2|C_1)\mathbb{P}(C_3|C_1 \cap C_2)$

- Law of Total Probability: Suppose k mutually exclusive and exhaustive events $C_1, C_2, \dots, C_k$ such that $\mathbb{P}(C_i) > 0$ and $\mathcal{C} = C_1 \cup C_2 \cup \dots \cup C_k$, we have

$$\mathbb{P}(C) = \sum_{i=1}^k \mathbb{P}(C_i) \mathbb{P}(C|C_i).$$

- Bayes's Theorem:

$$\mathbb{P}(C_j|C) = \frac{\mathbb{P}(C \cap C_j)}{\mathbb{P}(C)} = \frac{\mathbb{P}(C_j) \mathbb{P}(C|C_j)}{\sum_{i=1}^k \mathbb{P}(C_i) \mathbb{P}(C|C_i)}.$$

- Bowl C_1 —3 red and 7 blue chips; Bowl C_2 —8 red and 2 blue chips; All chips are identical in size and shape. A die is cast and C_1 is selected if 5 or 6 appear; otherwise C_2 is selected. Compute the conditional probability of bowl C_1 is selected given that a red chip is drawn.

$$\begin{aligned}\mathbb{P}(C_1|C) &= \frac{\mathbb{P}(C_1)\mathbb{P}(C|C_1)}{\mathbb{P}(C_1)\mathbb{P}(C|C_1) + \mathbb{P}(C_2)\mathbb{P}(C|C_2)} = \frac{\frac{2}{6} \frac{3}{10}}{\frac{2}{6} \frac{3}{10} + \frac{4}{6} \frac{8}{10}} \\ &= \frac{3}{19}\end{aligned}$$

- Independent: $\mathbb{P}(C_2|C_1) = \mathbb{P}(C_2)$.
- C_1 and C_2 are independent iff $\mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2)$.
- Theorem: If A and B are independent, then A and B^c are also independent.

- C_1 , C_2 and C_3 are mutually independent iff they are pairwise independent:

$$\mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2), \quad \mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2)$$

$$\mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2), \quad \mathbb{P}(C_1 \cap C_2 \cap C_3) = \mathbb{P}(C_1)\mathbb{P}(C_2)\mathbb{P}(C_3).$$

- Many combinations of these events and their complements are independent, such as
 - The events C_1^c and $C_2 \cap C_3^c \cap C_4$ are independent;
 - The events $C_1 \cup C_2^c$, C_3^c and $C_4 \cap C_5^c$ are mutually independent.
→ **show this as an exercise**